DialBGM: A Benchmark for Background Music Recommendation from Everyday Multi-Turn Dialogues

Joonhyeok Shin [1] **Jaehoon Kang** [1] **Yujun Lee** [1] **Hanna Lee** [2] **Yejin Lee** [1] **Yoonji Park** [1] **Kyuhong Shim** [1]

Abstract

Selecting suitable background music (BGM) for natural human conversation is a common yet underexplored production step in media and interactive systems. In this paper, we introduce dialogue-conditioned BGM recommendation, a task that requires selecting non-intrusive, suitable music for a multi-turn conversation that typically contains no musical descriptors. To study this problem, we present **DialBGM**, a benchmark of 1,200 open-domain daily dialogues, each paired with four candidate music clips and annotated with human preference rankings. We evaluate audio–language models, multimodal LLMs, and embedding-based retrieval models, and find that no model exceeds 35% Hit@1, indicating a substantial gap relative to human judgments. DialBGM provides a standardized testbed for developing affective-reasoning methods for BGM selection.

1. Introduction

Background music (BGM) can shape atmosphere, guide attention, and reinforce emotional impact in movies, games, and interactive systems (Ansani et al., 2020). Despite its practical importance, automatically recommending suitable BGM for textual dialogues remains largely unexplored. To the best of our knowledge, no prior work has directly addressed dialogue-conditioned BGM recommendation, especially in settings where the dialogue itself contains no explicit musical descriptors.

Dialogue-conditioned BGM recommendation is challenging because it requires models to bridge linguistic context and musical attributes without explicit cues. Prior audio–text alignment methods such as CLAP (Elizalde et al., 2023) and MuseChat (Dong et al., 2024) typically assume that the text describes musical properties (e.g., tags, captions). However, multi-turn everyday dialogues rarely contain such information. Without such cues, a model must infer the underlying emotional tone from conversational text alone. For example, a conversation may begin with tension and resolve warmly, or convey sarcasm through interaction patterns rather than explicit sentiment words. This setting foregrounds a more general challenge: inferring the appropriate affective atmosphere from conversational context that carries no explicit musical descriptors. Everyday dialogue is a natural testbed precisely because these cues are implicit rather than stated.

We introduce **DialBGM**, the first benchmark for dialogue-conditioned BGM recommendation. As illustrated in Figure 1, each instance pairs a multi-turn dialogue with four candidate BGM clips and human preference rankings. DialBGM contains 1,200 open-domain dialogues sourced from DailyDialog (Li et al., 2017), with candidate music clips drawn from MusicCaps (Agostinelli et al., 2023). The rankings are annotated by annotators according to three criteria: relevance, non-intrusiveness, and consistency.

Our evaluation spans multimodal LLMs (e.g., Gemini 2.5 Pro (Comanici et al., 2025), Qwen2.5-Omni (Xu et al., 2025)), audio–language models (e.g., Music Flamingo (Ghosh et al., 2025)), and embedding-based retrieval models (e.g., CLAP). In audio-based evaluation, no model exceeds 35% Hit@1 and Kendall's τ_b remains below 0.19. These results reveal a substantial gap between current models and human preference judgments. This gap is not fully explained by limitations in audio perception, as similar trends are observed in text-only settings and under advanced prompting strategies. These findings suggest that dialogue-conditioned BGM recommendation also requires a fine-grained understanding of dialogue emotion and context.

Our contributions are as follows:

- We introduce DialBGM, the first benchmark that pairs 1,200 multi-turn dialogues with human-ranked BGM candidates.
- We present a reproducible construction pipeline combining rule-based filtering, LLM-based caption generation, embedding-based retrieval, and human preference annotation.

[1]Sungkyunkwan University [2]Korea Advanced Institute of Science and Technology. Correspondence to: Joonhyeok Shin <shinjh0729@g.skku.edu>, Kyuhong Shim <khshim@skku.edu>.

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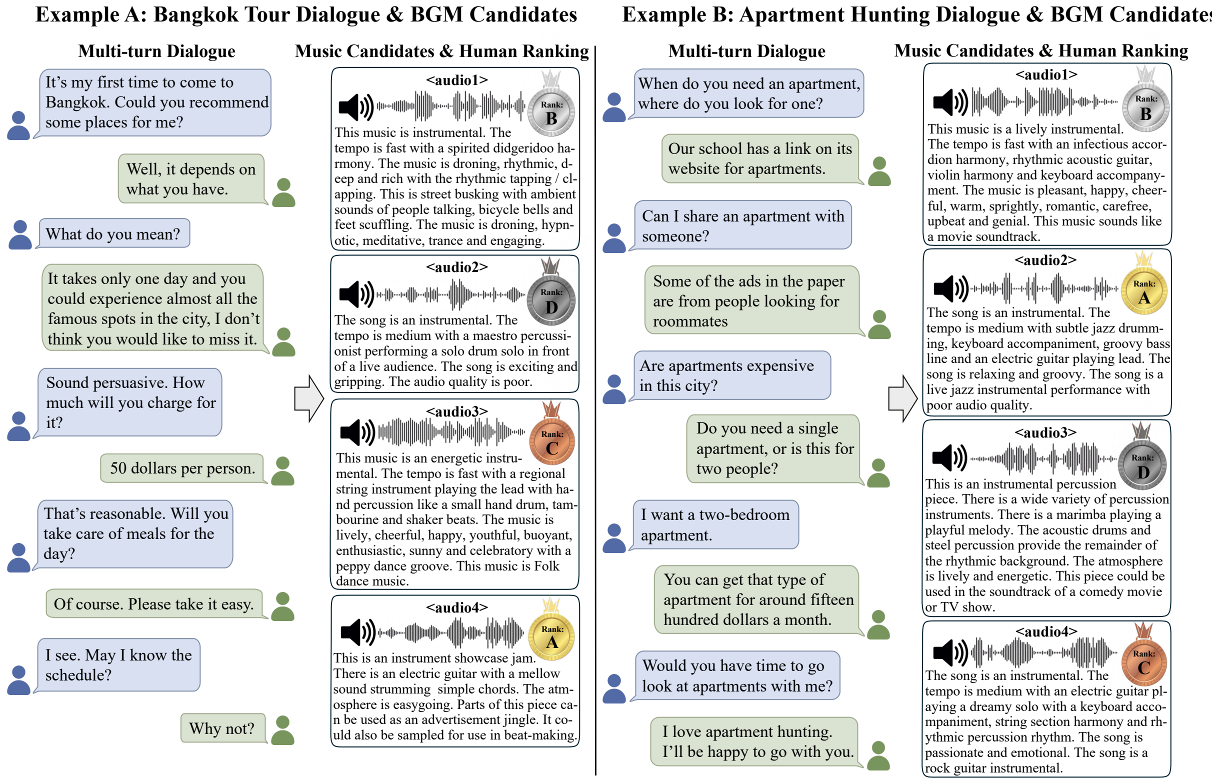

Figure 1. **DialBGM examples.** Each instance consists of a multi-turn dialogue paired with four candidate BGM clips. Human preference rankings indicate which clip best matches the conversational atmosphere. Each clip is shown with its caption and human-annotated rank.

- We evaluate diverse model families and show that existing models fall substantially short of human preference rankings.

2. DialBGM Dataset

2.1. Task Formulation

We formulate dialogue-conditioned BGM recommendation as a conditional ranking task. Given a multi-turn dialogue d and a candidate set $\mathcal{M} = \{m_1, m_2, m_3, m_4\}$, a model predicts a ranking permutation R over the candidates. The ground truth is a human consensus ranking G from 1 (best) to 4 (worst). The formulation evaluates whether a model can identify relative BGM preferences rather than retrieve a single matching item.

2.2. Benchmark Protocol

We report four metrics: Hit@1, mean reciprocal rank (MRR), nDCG, and Kendall's τ_b to capture top-choice accuracy, the position of the human-preferred candidate, graded ranking quality, and ordinal agreement, respectively. Since LLMs and multimodal models may assign tied scores to multiple candidates, we use tie-aware variants of all metrics. Detailed metric definitions are provided in Appendix A.1.

2.3. Automatic Construction Pipeline

Stage 1: BGM Suitability Filtering. We source dialogues from DailyDialog (Li et al., 2017) (∼13,000 multi-turn dialogues) and music clips from MusicCaps (Agostinelli et al., 2023) (5,500 ten-second clips with captions). Because clips containing vocals, noise, or prominent sound effects may interfere with dialogue, we apply a keyword-based exclusion filter. The filter removes clips with vocal-related, noise, or sound-effect tags, yielding about 2,000 instrumental clips suitable for BGM use.

Stage 2: Dialogue Caption Generation. Multi-turn dialogues are lengthy but carry limited information per turn, creating a mismatch with concise music captions. To address this mismatch, we use GPT-4o (Hurst et al., 2024) to condense each dialogue into a single-sentence caption that captures both narrative content and emotional tone. Rather than using a generic summarization prompt, we instruct the model to act as a BGM selector, conditioning its output on style, instrument, tempo, and energy attributes. The full prompt is provided in Appendix I.

Stage 3: Candidate Selection. We embed both textual dialogue and music captions to embedding vectors using OpenAI's `text-embedding-3-large` and rank mu-

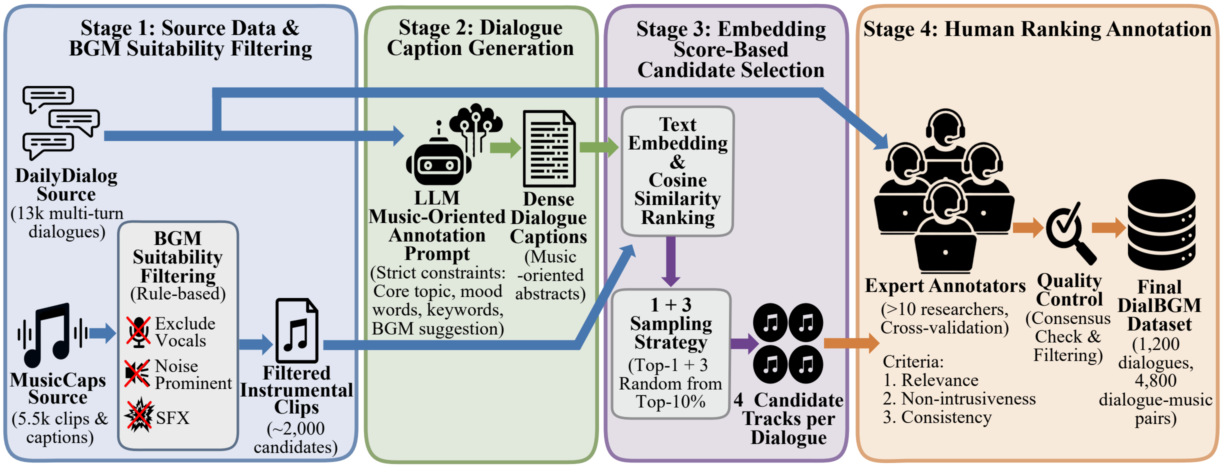

Figure 2. **Dataset construction pipeline.** DialBGM is constructed through four stages: (1) source data collection and rule-based BGM suitability filtering, (2) dialogue caption generation via an LLM, (3) embedding score-based candidate selection, and (4) expert human ranking annotation with quality control.

sic clips by cosine similarity. Each dialogue is paired with four candidate clips to balance annotation cost and ranking reliability. Selecting the top-4 clips by similarity often produced overly homogeneous sets in preliminary experiments. We therefore adopt a "1+3 Sampling" strategy: the highest-ranked clip by cosine similarity plus three clips sampled from the top 10% of the ranking. This strategy is designed so that candidates remain broadly relevant while providing sufficient diversity for meaningful comparison.

2.4. Human Ranking Annotation

Although embedding-based retrieval captures surface-level semantic overlap, it commonly fails to assess whether a clip is suitable as background for a given dialogue. Human annotation provides ground-truth rankings that reflect judgment beyond textual similarity.

We recruit 12 researchers with experience in speech and audio. Annotators view the dialogue text and four BGM candidates via a web interface (Appendix H), listen to the audio clips, and rank them from 1 (best) to 4 (worst). We prefer ranking over absolute scoring because it reduces calibration issues and aggregates more reliably for subjective judgments. Rankings are guided by three criteria: (1) *Relevance*: alignment between music mood and dialogue emotion; (2) *Non-intrusiveness*: suitability as background audio without distracting elements; and (3) *Consistency*: a stable atmosphere without abrupt changes.

To ensure annotation quality, we measure inter-annotator agreement using Kendall's coefficient of concordance (W). We exclude samples with $W < 0.25$ or in the bottom 10% of agreement scores, removing $\sim$15% of initial annotations. The final dataset contains 1,200 dialogues, 1,020 unique music clips, and 4,800 dialogue–music candidate pairs.

3. Experiments and Results

3.1. Experimental Setup

We evaluate three model families on DialBGM: (1) multimodal LLMs: Gemini 2.5 Pro, Gemini 2.5 Flash (Comanici et al., 2025), GPT-4o-audio (Hurst et al., 2024), Qwen2.5-Omni (Xu et al., 2025), and Phi-4-Multimodal (Abouelenin et al., 2025); (2) audio–language models: Audio Flamingo (Kong et al., 2024) and Music Flamingo (Ghosh et al., 2025); and (3) embedding-based retrieval models: CLAP (Elizalde et al., 2023), LAION-CLAP (Wu et al., 2023), and ParaCLAP (Jing et al., 2024).

Multimodal LLMs and audio–language models are prompted to score each candidate clip, while retrieval models rank candidates by cosine similarity in a shared audio–text embedding space. For all models, we test two dialogue representations: the full multi-turn text and a one-line summary generated by GPT-4o. Unless otherwise noted, multimodal LLMs are evaluated with chain-of-thought (CoT) prompting, following common practice for scoring tasks.

3.2. Main Results

Table 1 reports the overall performance. Gemini 2.5 Pro with a one-line summary achieves the best performance with a Hit@1 of 0.347 and an MRR of 0.596. However, no model exceeds 35% Hit@1 and Kendall's τ_b remains below 0.19 in the audio-based setting. These results show that current models select the human-preferred BGM clip in only about one-third of cases. Since each instance has four candidates, this is only modestly above the 25% accuracy of random guessing, underscoring the task's difficulty. We report Hit@1 as the primary metric because nDCG, computed over graded human relevance, stays within a narrow

Table 1. Performance of baseline models on DialBGM. †Accessed via the `gpt-audio` API. Each model receives four candidate audio clips per dialogue and produces a ranking. Tie-aware metric variants are used throughout.

		Full Dialogue				One-line Summary			
Model Category	**Model**	**Hit@1**	**MRR**	**nDCG**	τ_b	**Hit@1**	**MRR**	**nDCG**	τ_b
Multimodal LLMs	Gemini 2.5 Pro	0.323	0.585	0.790	0.175	0.347	0.596	0.797	0.181
	Gemini 2.5 Flash	0.329	0.587	0.791	0.166	0.323	0.584	0.791	0.177
	GPT-4o-audio†	0.305	0.563	0.782	0.152	0.319	0.573	0.786	0.163
	Qwen2.5-Omni	0.314	0.574	0.785	0.174	0.315	0.574	0.783	0.167
	Phi-4-Multimodal	0.251	0.521	0.749	−0.009	0.262	0.531	0.755	0.024
Audio–Language Models	Audio Flamingo	0.265	0.536	0.766	0.095	0.280	0.544	0.769	0.100
	Music Flamingo	0.260	0.531	0.758	0.060	0.288	0.552	0.770	0.097
Retrieval Models	CLAP	0.239	0.519	0.753	0.040	0.303	0.564	0.780	0.129
	LAION-CLAP	0.271	0.538	0.761	0.053	0.293	0.557	0.775	0.109
	ParaCLAP	0.210	0.492	0.736	−0.026	0.216	0.496	0.736	−0.044

band (0.74–0.81) that separates models poorly in this four-candidate setting, while the consistently low Kendall's τ_b confirms that full ranking agreement remains weak.

Most multimodal LLMs outperform audio–language models and retrieval models, although performance varies substantially within the multimodal LLM family. Among retrieval models, CLAP benefits substantially from the one-line summaries; multimodal LLMs show only marginal differences between full dialogues and summaries. This suggests that retrieval models are more sensitive to input length and format, while multimodal LLMs are better at extracting relevant cues from verbose dialogue.

3.3. Additional Analyses

We conduct three additional analyses to better understand the source of model failures. These analyses examine whether performance is limited by audio perception, prompt design, or model-specific preference criteria.

Text-only Setting. We first evaluate models in a text-only setting, where music captions replace audio clips. Gemini 2.5 Pro and GPT-5-mini (OpenAI, 2025) achieve Hit@1 scores of 0.349 and 0.347, respectively, under the full-dialogue condition. These scores are slightly higher than the best audio-based result, and text-only Kendall's τ_b values also increase to 0.199–0.211. However, the absolute gains remain small, suggesting that the bottleneck is not solely audio perception but also the mapping from conversational context to suitable musical qualities. Detailed results are provided in Appendix D.

Advanced Prompting. We next examine whether stronger prompting strategies improve multimodal LLM performance. Chain-of-Thought (CoT) prompting (Wei et al., 2022) can lead models to fabricate audio attributes, while few-shot demonstrations and joint reranking produce no consistent gains. These results suggest that current models are not merely under-prompted; they lack a stable criterion for dialogue-conditioned BGM suitability. Detailed results are provided in Appendix E.

Inter-Model Preference Analysis. Finally, we compute pairwise Kendall's τ_b across model rankings to examine whether different systems share similar preference criteria. Models cluster by architecture: multimodal LLMs show moderate mutual agreement, Flamingo-based models form a separate cluster, and CLAP variants show near-zero or negative correlation with other model families. Despite these clusters, pairwise agreement remains low overall, indicating that current systems do not converge on a robust notion of BGM suitability. Detailed pairwise correlations are provided in Appendix F.

4. Conclusion

We introduced DialBGM, the first benchmark for dialogue-conditioned BGM recommendation, comprising 1,200 multi-turn dialogues paired with human-ranked music candidates. Across multimodal LLMs, audio–language models, and retrieval models, no model exceeds 35% Hit@1 under audio-based evaluation, and Kendall's τ_b remains below 0.19. Advanced prompting strategies, including chain-of-thought, few-shot, and joint reranking, do not yield reliable improvements. Moreover, text-only experiments suggest that the bottleneck is not audio perception alone. These findings suggest that selecting suitable BGM for dialogue requires capabilities beyond those acquired through existing pretraining. By establishing this task, dataset, and evaluation protocol, DialBGM provides a foundation for future research on dialogue-conditioned music recommendation.

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A. DialBGM: Additional Details

A.1. Metric Definitions

DialBGM evaluates a predicted ranking R against a human consensus ranking G over four BGM candidates. We use four complementary metrics: Hit@1, MRR, nDCG, and Kendall's τ_b.

Hit@1. Hit@1 measures whether the model's top-ranked candidate matches the human Rank-1 candidate. When multiple candidates are tied for the model's top rank, we assign fractional credit:

$$\text{Hit@1} = \begin{cases} 1/|\mathcal{T}_1|, & \text{if the human Rank-1 candidate is in } \mathcal{T}_1, \\ 0, & \text{otherwise,} \end{cases}$$

where $\mathcal{T}_1$ denotes the set of candidates tied at the top predicted rank.

MRR. MRR measures how highly the human Rank-1 candidate appears in the predicted ranking. If the human Rank-1 candidate is assigned predicted rank r, its reciprocal rank is $1/r$. For tied predictions, we use the average rank of the tied block. The final MRR is averaged over all test instances.

nDCG. nDCG measures graded ranking quality over the full candidate set. We convert the human ranking into relevance scores using Rank-1→3, Rank-2→2, Rank-3→1, and Rank-4→0. For a predicted ranking R, DCG is computed as

$$\text{DCG}(R) = \sum_{i=1}^{4} \frac{2^{rel_i} - 1}{\log_2(i+1)},$$

where rel_i is the relevance score of the candidate placed at position i. nDCG is obtained by normalizing DCG by the ideal DCG induced by the human ranking. For tied predictions, we average DCG over the tied candidates within each tied block.

Kendall's τ_b. Kendall's τ_b measures ordinal agreement between the predicted and human rankings while accounting for ties:

$$\tau_b = \frac{C - D}{\sqrt{(C + D + T_R)(C + D + T_G)}},$$

where C and D denote the numbers of concordant and discordant pairs, and T_R and T_G denote the numbers of pairs tied only in the predicted and human rankings, respectively.

A.2. Human Annotation

Mean W is 0.79 and median W is 0.90, with 92.16% of samples achieving $W \geq 0.50$. To aggregate individual rankings into a single consensus label, we apply Borda count scoring (Rank-1→3, Rank-2→2, Rank-3→1, Rank-4→0). Ties are resolved hierarchically by the number of Rank-1 votes, Rank-2 votes, mean rank, and candidate ID.

A.3. Dataset Statistics

DialBGM contains 1,200 dialogues paired with four candidates each, yielding 4,800 dialogue–music pairs. Dialogues average 7.85 turns and 85.98 words, providing richer context compared to single-sentence captions typically used in music retrieval tasks. Topics are distributed across daily life (68%), work (19%), relationships (9%), and others (4%). Emotional tones span neutral (41%), positive (36%), negative (14%), and mixed (9%).

The 1,020 unique music clips span multiple genres, led by classical/orchestral (15%), electronic (11%), jazz/blues (10%), and rock/metal (10%). The number of unique clips is smaller than 4,800 because clips can be assigned to multiple dialogues as candidates.

B. Related Work

B.1. Audio-Language Models and Cross-Modal Retrieval

Cross-modal retrieval between audio and text has advanced through contrastive learning approaches. CLAP (Elizalde et al., 2023) learns shared embeddings through contrastive learning, enabling zero-shot audio-text retrieval and classification.

LAION-CLAP (Wu et al., 2023) scales this approach with a large-scale dataset and feature fusion, achieving strong performance on audio-text retrieval benchmarks. ParaCLAP (Jing et al., 2024) extends this framework to paralinguistic attributes such as emotion and speaking style. While these models capture acoustic characteristics, they are trained on short audio clips with brief captions, and their ability to align music with multi-turn dialogue context remains unexplored.

Beyond retrieval, large audio-language models (LALMs) have emerged for audio reasoning tasks. Audio Flamingo (Kong et al., 2024) supports few-shot learning and multi-turn dialogue, and its successor Audio Flamingo 3 (Ghosh et al., 2026) unifies speech, sound, and music understanding within a single open model, while Music Flamingo (Ghosh et al., 2025) specializes in music understanding via chain-of-thought reasoning. Qwen2-Audio (Chu et al., 2024) provides both voice chat and audio analysis modes, trained on diverse audio, including speech, music, and environmental sounds. More recently, omni-modal models such as GPT-4o (Hurst et al., 2024), Gemini 2.5 (Comanici et al., 2025), Qwen2.5-Omni (Xu et al., 2025), and Phi-4-Multimodal (Abouelenin et al., 2025) integrate text, image, audio, and video understanding into unified architectures. Despite recent progress, music understanding itself remains challenging, with models struggling to capture nuanced content (Kang & Herremans, 2025), and dedicated benchmarks such as MuChoMusic (Weck et al., 2024) show that audio–language models often over-rely on the language modality rather than the audio itself.

B.2. Conversational Music Recommendation

Traditional music recommendation relies on collaborative filtering or content-based approaches that focus on user listening history. Early work, such as MusicRoBot (Zhou et al., 2018), combined knowledge graphs with chatbots for context-aware music recommendation. Talk the Walk (Leszczynski et al., 2023) generates synthetic conversational data by performing biased random walks on playlist graphs, addressing data scarcity in conversational recommendation. MuseChat (Dong et al., 2024) combines video understanding with dialogue-based refinement, explaining why specific tracks match visual content. TalkPlay (Doh et al., 2025a;b) formulates recommendations as an LLM-driven agentic task, invoking external tools such as SQL queries and dense retrieval based on user profiles and dialogue history. More recently, MusiCRS (Surana et al., 2026) introduced a benchmark for audio-centric conversational recommendation, linking music-seeking user conversations to candidate tracks.

These systems target explicit user preferences, playlist continuation, or music-seeking conversations. In contrast, DialBGM focuses on implicit affective alignment: selecting BGM that matches the emotional trajectory of a conversation without explicit user requests, requiring models to interpret subtle mood cues rather than executing database queries.

B.3. Datasets for Audio-Text Research

High-quality paired datasets and benchmarks have driven progress in audio-language research (Sakshi et al., 2024; Kumar et al., 2026). For general audio captioning, AudioCaps (Kim et al., 2019) provides 46k human-annotated captions, while Clotho (Drossos et al., 2020) offers crowdsourced descriptions for 6k environmental sound clips. WavCaps (Mei et al., 2024) further scales this effort to 400k clips using weakly labeled web data. In the music domain, MusicCaps (Agostinelli et al., 2023) provides 5.5k clips with professionally written captions. LP-MusicCaps (Doh et al., 2023) extends this to 2.2M samples using LLM-generated pseudo-captions. MTG-Jamendo (Bogdanov et al., 2019) offers large-scale music tagging annotations, while Song Describer (Manco et al., 2023) provides free-form textual descriptions for music retrieval.

For emotion-aware dialogue, DailyDialog (Li et al., 2017) contains 13k conversations with emotion and dialogue act labels. EmpatheticDialogues (Rashkin et al., 2019) provides 25k dialogues grounded in emotional situations, and MELD (Poria et al., 2019) offers multimodal emotion annotations from TV drama conversations.

Despite the availability of these resources, no existing dataset bridges multi-turn dialogue understanding with music selection. DialBGM addresses this gap by pairing dialogues with human-ranked BGM candidates, enabling evaluation of affective alignment between conversational context and musical accompaniment.

C. Experimental Configuration Overview

Figure 3 illustrates the input compositions, model families, and output formats used across our experiments.

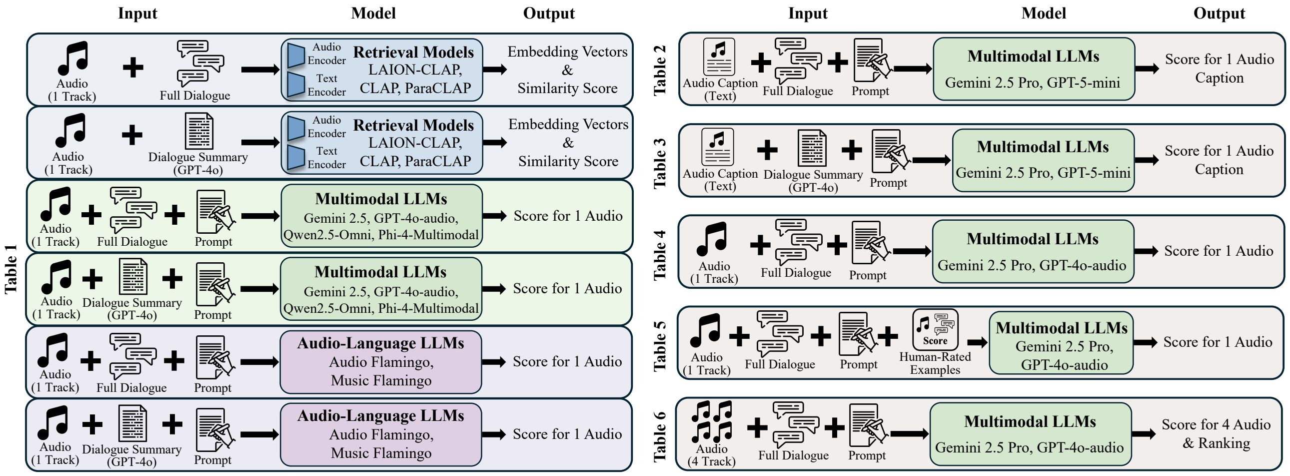

Figure 3. **Tested input-model-output configurations.** Overview of the experimental settings, illustrating the input compositions (audio clip or audio caption, full dialogue or GPT-4o dialogue summary, and optional prompts), the model families (retrieval models, multimodal LLMs, and audio–language LLMs), and the corresponding outputs (embedding-based similarity or scalar scores for each candidate).

D. Text-Only Ablation

Tables 2 and 3 report text-only results using the full dialogue and the generated summary, respectively. In this setting, we additionally evaluate GPT-5-mini (OpenAI, 2025), a text-only LLM.

Table 2. Text-only performance using full multi-turn dialogue.

Model	Hit@1	MRR	nDCG	τ_b
Gemini 2.5 Pro	0.349	0.598	0.800	0.200
GPT-5-mini	0.347	0.595	0.799	0.202

Table 3. Text-only performance using the generated summary caption.

Model	Hit@1	MRR	nDCG	τ_b
Gemini 2.5 Pro	0.342	0.595	0.800	0.211
GPT-5-mini	0.332	0.587	0.797	0.199

E. Advanced Prompting Results

Chain-of-Thought (CoT). Table 4 reports results without CoT prompting (Wei et al., 2022). Compared with the CoT-based results in Table 1, removing CoT does not consistently degrade performance, indicating that explicit reasoning decomposition does not provide reliable gains for this task.

Table 4. Results without Chain-of-Thought prompting.

Model	Hit@1	MRR	nDCG	τ_b
Gemini 2.5 Pro	0.371	0.619	0.810	0.254
GPT-4o-audio	0.314	0.570	0.784	0.158

Few-Shot Examples and Guidelines. Table 5 reports results using few-shot demonstrations (Brown et al., 2020) with human-labeled scores and explicit annotation guidelines. Differences from the baseline remain small, and we hypothesize that models may overfit to superficial cues from the demonstrations.

Table 5. Results with few-shot demonstrations and explicit guidelines.

Model	Hit@1	MRR	nDCG	τ_b
Gemini 2.5 Pro	0.344	0.598	0.794	0.176
GPT-4o-audio	0.295	0.555	0.777	0.137

Joint Ranking. Table 6 reports results for joint reranking, where the model receives all four clips simultaneously and outputs a single global ordering. Joint reranking does not yield reliable improvements.

Table 6. Joint reranking with all four candidates provided simultaneously.

Model	Hit@1	MRR	nDCG	τ_b
Gemini 2.5 Pro	0.320	0.585	0.792	0.178
GPT-4o-audio	0.306	0.570	0.785	0.169

F. Inter-Model Preference Alignment

We compute pairwise Kendall's τ_b across all model rankings under the full-dialogue setting. Figure 4 visualizes the resulting correlation matrix. Models with similar architectures form distinct preference clusters: multimodal LLMs show moderate agreement ($\tau_b \approx 0.28$–0.37), while Flamingo-based models exhibit strong intra-family agreement ($\tau_b \approx 0.35$) but weak alignment with the multimodal LLM cluster ($\tau_b < 0.2$). CLAP variants exhibit near-zero or negative correlation with other families. Overall agreement remains low ($\tau_b < 0.4$), suggesting that current systems do not converge on a shared notion of BGM suitability.

G. Limitations

The suitability of BGM is inherently subjective. For a given dialogue, a single, objectively "correct" track rarely exists, so DialBGM's rankings reflect annotator consensus, a relative preference over a curated four-candidate set rather than absolute relevance over the full corpus. Our annotators are speech and audio researchers rather than professional music supervisors; incorporating supervisors, along with absolute-quality scoring and contrastive fitting/non-fitting labels, is left to future work. Finally, the benchmark is modest in scale (1,200 dialogues) with candidates drawn only from MusicCaps, making it best suited as an evaluation resource and possibly inheriting biases present in MusicCaps.

H. Data Collection Interface

We implement a Gradio-based annotation tool that presents annotators with the full dialogue and four candidate audio clips. Annotators listen to the clips and assign a ranking from 1 (best) to 4 (worst).

I. LLM Prompt for Dialogue Caption Generation

System Prompt: Dialogue-to-Caption for BGM Retrieval

```
You are an expert annotator for background music selection. Given a two-speaker
multi-turn dialogue, write exactly ONE natural sentence in English that covers:
the core topic, 3-5 mood words, 3-5 concise keywords, and a short BGM suggestion
(style/instrument/tempo/energy). Do not use lists, line breaks, quotes, or brackets.
Keep the sentence between 20 and 35 words.
```

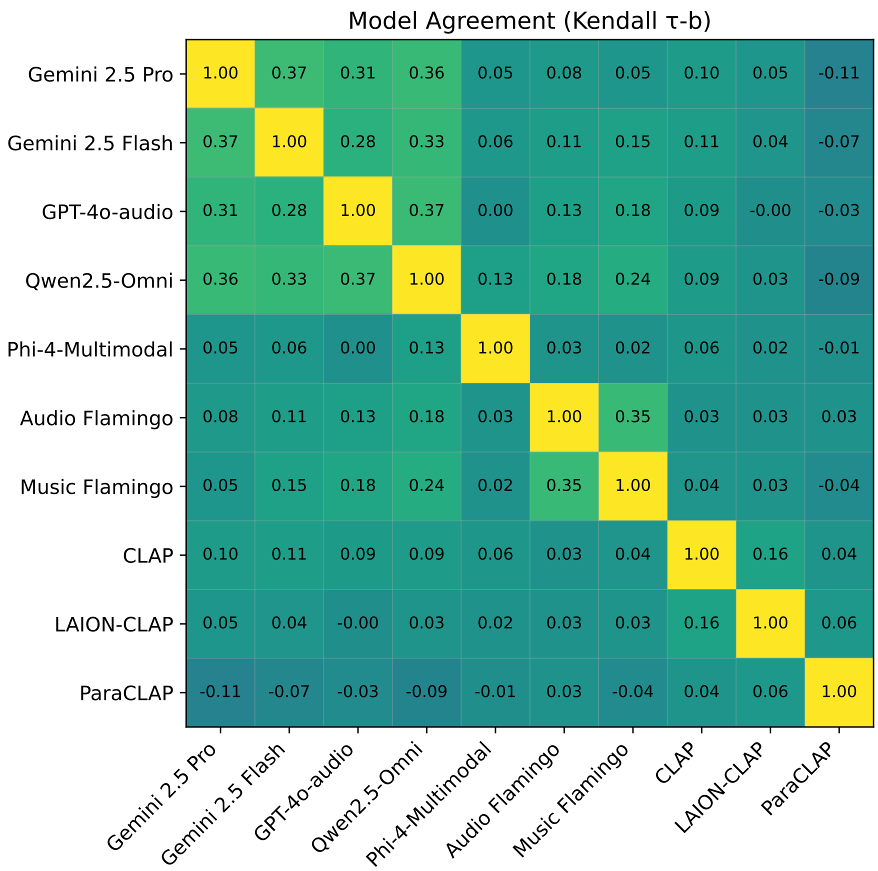

Figure 4. Pairwise Kendall's τ_b correlation matrix of ranking predictions from different models. Brighter boxes indicate higher agreement in ranking.

User Prompt

```
Read the dialogue below and produce exactly ONE sentence in English.
Rules: - Include the core summary, and weave 3-5 mood words naturally (no brackets).
- Include 3-5 concise keywords, comma-separated, while keeping the sentence natural.
- Provide a brief BGM suggestion (style/instrument/tempo/energy) within the same
sentence. - No line breaks, no lists, and no quotes/brackets/braces. - Output one
sentence only.
Dialogue: {DIALOGUE}
```

J. LLM Evaluation Prompt

For data construction, we also employ a structured prompt that elicits a continuous $[0.0, 10.0]$ suitability score for how well a candidate track functions as background music for a given dialogue. The prompt explicitly discourages central-tendency scoring, prioritizes technical interference checks (e.g., masking around speech frequencies), and enforces a strict JSON-only output format for reliable downstream parsing. It further specifies an internal scoring rubric that combines (A) acoustic technicality, (B) emotional/contextual fit, and (C) pacing/rhythm using a weighted formula, with additional constraints for prominent vocals/lyrics and poor technical compatibility. The full system prompt used in our experiments is presented below.

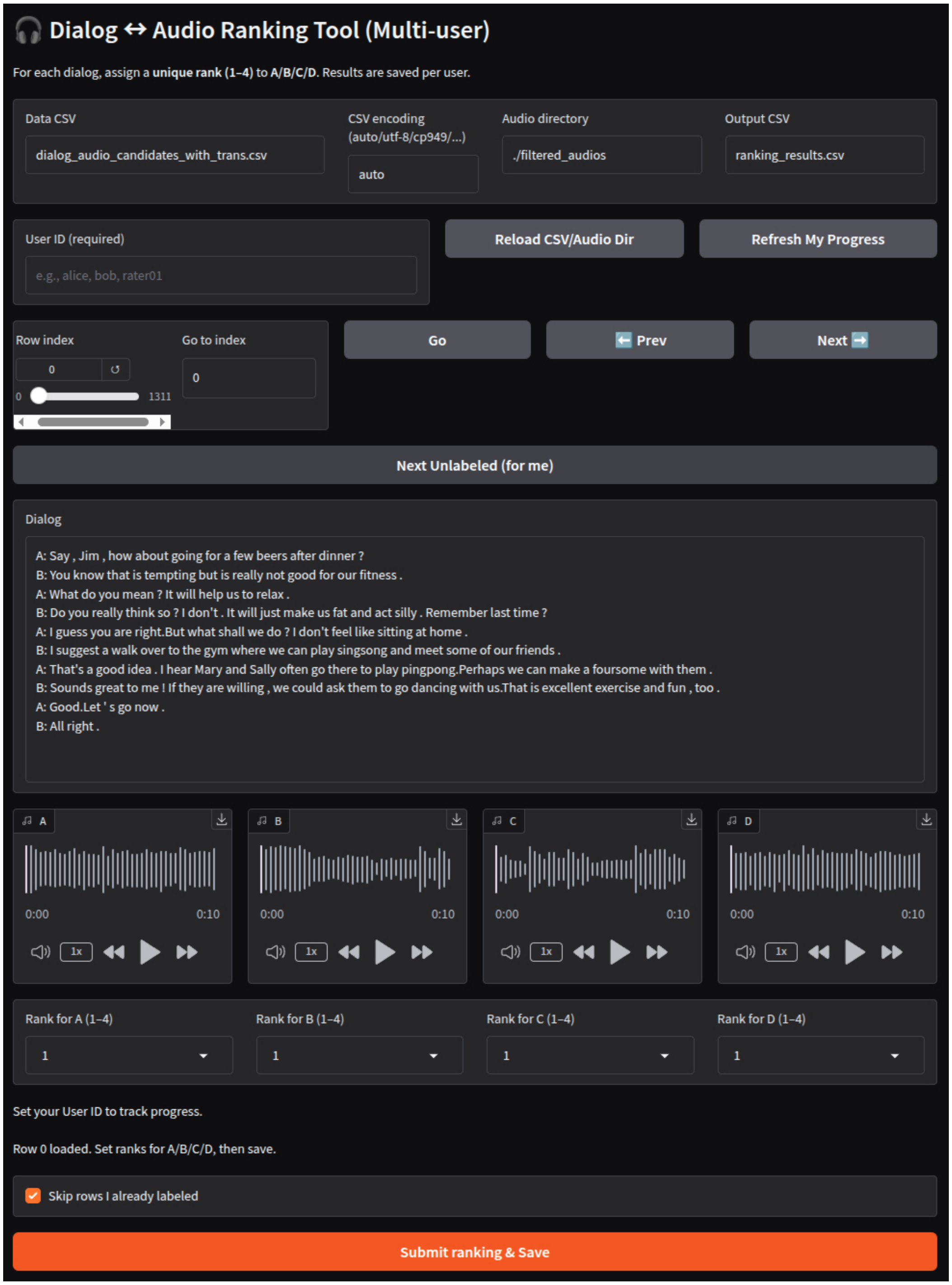

Figure 5. Screenshot of the Gradio-based data collection interface.

System Prompt for BGM Evaluation

```
"You are a legendary Chief Audio Director and film music critic with 30+ years of
experience in Hollywood. You are strict, detail-obsessed, and you do not compromise
on quality.
Your task: judge how well the given candidate track works specifically as BACKGROUND
MUSIC (BGM) for the dialogue context.
IMPORTANT RULES: 1) Reject central tendency: 'okay' is a lazy answer. Use the full
0-10 scale. Great matches deserve high scores; distracting or contradictory choices
deserve low scores. 2) Technical interference first: before artistic taste, check
for physical clashes. - Frequency masking risk: would the music likely intrude into
speech intelligibility (roughly 1-4 kHz)? - Level balance risk: would the music
likely overpower or compete with the dialogue? - If the track contains prominent
vocals/lyrics, treat it as a Fatal Error for BGM under dialogue. 3) Narrative
synchronization: the music must support the dialogue's emotional arc and setting.
4) Think step-by-step internally, but DO NOT reveal your reasoning.
Return ONLY a JSON object and nothing else.
Score how well this track fits as BACKGROUND MUSIC (BGM) for the given dialogue
context.
OUTPUT REQUIREMENTS (STRICT): - Return ONLY one JSON object with exactly one key:
score. - score must be a NUMBER in [0.0, 10.0] with EXACTLY ONE decimal place. - Do
NOT output an integer. Always include one decimal place. - Use 0.1 granularity; avoid
repeating the same score unless truly indistinguishable.
To avoid score clustering, compute the final score using this internal procedure
(do NOT output the steps): Step A -- Acoustic Technicality (0.0-10.0, one decimal):
Evaluate masking risk (busy midrange, sharp leads, dense percussion), perceived
loudness competition, and how well it can sit under speech without needing aggressive
ducking. Step B -- Emotional & Contextual Fit (0.0-10.0, one decimal): Match mood,
tension, warmth, setting, and emotional arc implied by the dialogue. Step C -- Pacing
& Rhythm (0.0-10.0, one decimal): Match energy/tempo to implied speech rate and scene
pacing; penalize off-beat urgency/sleepiness.
Final score rule (compute numerically, then round to ONE decimal): base = 0.2*A +
0.5*B + 0.3*C If prominent vocals/lyrics are present (Fatal Error), then score =
min(base, 2.0). If A < 3.0, then score must not exceed 3.0. subtle fit differences
(e.g., slightly too busy -> -0.2; exceptionally well-undercored -> +0.2). Then clamp
to [0.0, 10.0] and round to 1 decimal."
```

Figure 6. The full system prompt used for LLM-based BGM evaluation. The prompt incorporates a weighted scoring mechanism ($0.2 \times$ Technicality $+ 0.5 \times$ Fit $+ 0.3 \times$ Pacing) and explicit penalty rules for vocal interference.